

Executive Summary – Measure Phase

Project Title

Reducing Checkout Waiting Time at Campus Coffee Shop

Simple Project Overview

This project focuses on reducing how long students wait in line to pay at the campus coffee shop. During busy mornings, the line is long and many students either complain or walk away without buying. In the Measure phase, the team mapped the current process, agreed on clear definitions, built a data collection plan, and collected baseline data on waiting times and common delay reasons.

Summary of Key Deliverables

Process Maps

The process map shows each step from a student joining the line to leaving the counter with a drink: join queue, place order, pay, wait for drink, receive drink. The map helped the team see where the line pauses and where steps pile up on one staff member.

Operational Definitions

The team agreed on simple, clear definitions. “Checkout waiting time” is measured from the moment a student joins the line to the moment payment is complete. A “long wait” was defined as more than 7 minutes. These shared definitions reduced arguments and made the data easier to trust.

Data Collection Plan

The team decided to measure waiting time for every 3rd customer during the morning rush (8–10 a.m.) for two weeks. Cashiers used a simple timing sheet with start and end times. This plan also logged main delay reasons, like “card payment issue” or “barista waiting on supplies.”

Pareto Charts

The Pareto chart showed that a small number of causes explained most long waits: (1) only one cashier open, (2) price lookups for custom drinks, (3) slow card payments. These three reasons made up most of the total delay time.

VOP Matrix

The Voice of the Process (VOP) matrix linked process steps (queue, order, pay, pickup) with key output measures like wait time and abandoned orders. This made it clear that checkout and payment steps drive most pain for both students and staff.

VOC Results

Short student surveys and quick interviews showed that most students are okay with a 5-minute wait, but get frustrated after 7 minutes. Several said they would come less often if the line stayed slow. Staff also shared that mornings feel stressful and rushed.

VOP Baseline Results

Baseline data showed an average checkout waiting time of about 8.5 minutes during the morning rush, with roughly 35% of customers waiting more than 7 minutes. A few peak days were even worse. This confirmed the problem is real and not just based on complaints.

Basic Statistics

The team collected 200 usable waiting time data points. The average (mean) wait was around 8.5 minutes, the median was 8 minutes, and the standard deviation was about 2.3 minutes. This means waits are not only long, but also quite variable.

Gage R&R (Measurement System)

The team checked whether different cashiers recorded waiting time in a similar way. Two cashiers timed the same small set of customers and their numbers were very close. Because the timing was simple (start and stop time with one clock), measurement error looked small and acceptable.

Process Capability (if treated as a “time under 7 minutes” spec)

If the “specification” is that customers should wait less than 7 minutes to pay, the current process capability is poor. Many data points are above 7 minutes, which matches what students say. This supports the need for improvement.

Summary of Next Steps

Next, the team will move to the Analyze phase. The main focus will be to understand why only one cashier is open, why price lookups are slow, and what is causing card payment delays. Simple tests will be used to see which ideas reduce waiting time the most without adding too much cost or complexity.

Overarching Questions

a) Precisely define what problems are occurring.

Students are waiting too long to pay at the coffee shop during the morning rush. Average checkout waiting time is about 8.5 minutes, and more than a third of customers wait over 7 minutes. Some students leave the line without buying anything, and both staff and customers report stress and frustration.

b) How did you prioritize and select critical input, process, and output measures?

The team started from the customer view: students mainly care about how long they wait. So the main output measure is checkout waiting time in minutes. Input and process measures were chosen by asking “what slows the line down?” and by talking with staff: number of open cash registers, number of people in line, payment method (cash vs card), and whether staff had to do price lookups. These measures are easy to collect and clearly linked to the wait time.

c) What have you done to validate the measurement system?

The team kept the measurement system very simple: one clock and one form. Cashiers practiced recording start and end times and compared results on a short trial. Two cashiers timed the same customers and got almost the same values. This quick check suggested that the measurement system is consistent enough for this project.

d) What patterns are exhibited in the data?

Wait times are usually shorter at the very start and end of the rush period, and longest around 8:30–9:00 a.m. Waits are much worse on Mondays and right after holidays. Most long waits are linked to times when only one cashier is active or when there is a run of complex orders that require manual price lookups.

e) Do you believe there was an appropriate use of tools?

Yes. For a simple campus shop, the team used just enough tools to understand the problem: a clear process map, simple operational definitions, a basic data collection plan, Pareto charts, and basic statistics. Nothing feels over-engineered, and every tool added some insight.

f) What tools did you use beyond each required tool? Why? What benefit did you gain?

Beyond the required tools, the team used: a short customer survey (VOC), a quick staff focus group, and a simple run chart of daily average wait times. The surveys and focus group helped confirm what really matters to people, while the run chart made it easy to see which days were worst and whether things were getting better or worse over time.

g) Do you believe the project is ready to move to the next phase? Why or why not?

Yes. The team has a good picture of how long customers are waiting, when waits are worst, and what main things seem to drive delays. The measurement system is simple and consistent, and there is enough data to start asking “why” and looking for root causes in the Analyze phase.

h) If the project is not ready, what measures need to be taken to recover?

At this point the project does look ready. If more work were needed, the team would collect one more week of data (especially after small process changes), double-check the timing method with all cashiers, and make sure the process map is still accurate when new staff are working.

i) Does the project charter, problem, scope, or other aspect of the project need to be refined?

A small refinement is helpful: clearly state that the focus is the checkout step (from joining the line until payment is done) during weekday mornings, not the full drink-making process. This sharper scope keeps the team focused and avoids trying to fix everything in the shop at once.